

	FIR VEHICLE TESTING		16/03/2026
FIR No :FIR/Testing-Outdoor-UA/518100/2026/01335			
Part No/ Group :51810368R			
Subject :Report on fluid economy evaluation of Signa 3023.T BSVI-II Karnali Ashwamedh 5.6L (220 HP) engine, G750 (FGR:8.45) Transmission and 5.12 Rear axle ratio in comparison with Signa 2823.T BSVI-II vehicle as a Base on JSR-Ranchi highway			
Test Vehicle	Project Name :Signa 3023.T BSVI-II Karnali Model Name : Signa 3023.T BSVI-II Karnali		Enclosures : Annexure 1 - 6
	Chasis Number :PC03505022RJ00755 Engine Number :12L63889392		
FIR Status :	ESO		

Role	Name	Signature
Prepared By	Palash Das	 15-Mar-2026
Checked By	Swapnil Jadhav	 16-Mar-2026
Approved By	Ajaykumar Jadhav	 16-Mar-2026

Cc: Head CVBU Engg, Chief Engineer, Head COC, Project Team [Design]; Project Team [Testing]

PA20 F222 (R4) March, 2021

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Documents : Please click here

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 FIR/Testing-Outdoor-UA/518100/2026/01335/A		FIR VEHICLE TESTING		Date: 09/03/2026			
Part No / Group: 51810368R		Specification no :- " 259051810029 "			<table><tr><td>Action by</td><td>Failure class</td></tr></table>	Action by	Failure class
Action by	Failure class						

Subject: -	Report on fluid economy evaluation of Signa 3023.T BSVI-II Karnali Ashwamedh 5.6L (220 HP) engine, G750 (FGR:8.45) Transmission and 5.12 Rear axle ratio in comparison with Signa 2823.T BSVI-II vehicle as a Base on JSR-Ranchi highway		
1.0 <u>Conclusion:</u> 1.1 <u>Fluid economy in laden(rated load) condition at 40 kmph :</u> - Fluid economy of Signa 3023.T Karnali BSVI-II vehicle was found 3.83 kmpl which was inferior by 4.1 % wrt Signa 2823.T BSVI-II Base vehicle. 1.2 <u>Fluid economy in laden(rated load) condition at 55 kmph :</u> - Fluid economy of Signa 3023.T Karnali BSVI-II vehicle was found 3.49 kmpl which was inferior by 5.0 % wrt Signa 2823.T BSVI-II Base vehicle. 1.3 <u>Fluid economy in Unladen condition at 40 kmph :</u> - Fluid economy of Signa 3023.T Karnali BSVI-II vehicle was found 6.84 kmpl which was Superior by 1.1 % wrt Signa 2823.T BSVI-II Base vehicle 1.4 <u>Fluid economy in Unladen condition at 55 kmph :</u> - Fluid economy of Signa 3023.T Karnali BSVI-II vehicle was found 6.19 kmpl which was inferior by 0.3% (At Par) wrt Signa 2823.T BSVI-II Base vehicle 1.5 <u>Overall fluid economy status at 40 kmph (80% Laden + 20% unladen condition):</u> - Fluid economy of Signa 3023.T Karnali BSVI-II vehicle was found 4.2 kmpl which was inferior by 3.45 % wrt Signa 2823.T BSVI-II Base vehicle - Signa 3023.T Karnali BSVI-II not meets the PALS target of "Leadership". 1.6 <u>Overall fluid economy status at 55 kmph (80% Laden + 20% unladen condition):</u> - Fluid economy of Signa 3023.T Karnali BSVI-II vehicle was found 3.83 kmpl which was inferior by 4.41 % wrt Signa 2823.T BSVI-II Base vehicle <u>Observation:</u> <u>Fluid Economy results by L/Ton/100 km:</u> ➤ <u>55 kmph cruising speed:</u> Average Fluid Economy of Signa 3023.T BSVI-II test vehicle was found Superior by 1.77% wrt Signa 2823.T BSVI-II base vehicle by L/Ton/100 km. ➤ <u>40 kmph cruising speed:</u> Average Fluid Economy of Signa 3023.T BSVI-II test vehicle was found Superior by 2.69% wrt Signa 2823.T BSVI-II base vehicle by L/Ton/100 km.		VI/ ENG -- VI/ ENG	C -- C

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2.0 Result: Vide Observations

FAILURES CLASS:

- A: Failure that happen without warning. Detrimental to vehicle / personnel, road users and safety. Can cause death / hospitalization. Noncompliance of regulatory requirement.
- B: Vehicle gets crippled with sufficient warning. Minor injury to occupant & road users (First aid only). Premature failures / repeated failures. Leads to consequential catastrophic damages. Need immediate attention.
- C: Customer irritants. Capable of damaging aesthetics. Failure can be attended at next servicing Schedule.
- D: Customer perception issues and minor irritants.

3.0 Test Vehicles: -

1. Signa 3023.T BSVI-II (ERCJ-1219)_Test vehicle
Chassis no. - PC03505022RJ00755; Engine no. - 12L63889392.
2. Signa 2823.T BSVI-II (ERCJ 1279)_Base vehicle
Chassis no. - PC03501122RJ00833; Engine no. - 22H63914054.

Enclosures:

- Annexure 1: Panel chart
Annexure 2: PAT sheet
Annexure 3: Duty cycle report
Annexure 4: Test request
Annexure 5: Test plan
Annexure 6: EOL

Input Doc Ref.: - Test Request No :- Testing-Outdoor-UA/03069
DID No :- NA.
DDP No: - NA.

Work Order No. :- P.21.EA.460	Make :- TATA Motors Limited	T I No. – UA003040
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Earlier Ref: - Nil.

4.0 Project Status:- DR3

5.0 Introduction:

As a part of Karnali vehicle Validation Program, Signa 3023.T Karnali vehicle fitted with M/s Cummins make Ashwamedh 5.6L BSVI-II (220 HP) engine, G750 (FGR:8.45) Transmission and 5.12 Rear axle ratio was offered for fluid economy evaluation along with Signa 2823.T BSVI-II vehicle as a base vehicle. All vehicles were tested back-to-back at JSR- Ranchi-JSR route in laden and unladen conditions at 40 km/hr as well as 55 km/hr vehicle cruising speeds. The subject report will provide detail about fluid economy evaluation test carried out on all vehicles (Test vehicle, Base vehicle).

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6.0 Testing:

Following test was conducted during evaluation :

Sr. no.	Test	Reference standard	Instruments used
1	Fluid economy evaluation	TST/VT/WG/158	Cummins Data Logger

7.0 Vehicle details:

Parameter		Base Vehicle	Test vehicle
Vehicle model		Signa 2823.T (OBD-I) (ERC-1279)	Signa 3023.T (OBD-II) (ERC-1219)
Engine Model		Cummins Ashwamedh 5.6L BS-VI (220 HP) OBD-II	Cummins Ashwamedh 5.6L BS-VI (220 HP) OBD-II
Max Power		220 HP @2300 RPM	220 HP @2300 RPM
Max Torque		925 Nm @ 1000 ~1300 rpm	925 Nm @ 1000 ~1300 rpm
GVW (kg)		28000	30000
Gear Box		Model: Tata G750, Type: DD	Model: Tata G750, Type: DD
Gear Ratios.		1st- 8.45, 2nd- 4.58, 3rd- 2.73, 4th- 1.78, 5th- 1.28, 6th- 1.00, Rev- 7.36	1st- 8.45, 2nd- 4.58, 3rd- 2.73, 4th- 1.78, 5th- 1.28, 6th- 1.00, Rev- 7.36
Axle Ratio		5.12	5.12
Tyres	FA1	Size: : 295/90R20, Make: M/s Ceat, Type : LCRR (Phase-II)	Size: : 295/90R20, Make: M/s Ceat, Type : LCRR (Phase-I) AW.
	RA1	Size: : 295/90R20, Make: M/s Ceat, Type : LCRR (Phase-II)	Size: : 295/90R20, Make: M/s Ceat, Type : LCRR (Phase-II)
	RA2	Size: : 295/90R20, Make: M/s Ceat, Type : LCRR (Phase-II)	Size: : 295/90R20, Make: M/s Ceat, Type : LCRR (Phase-II)
Tractive effort @ 1 st gear		77034 N	77034 N
Tractive effort @ Top gear		9116 N	9116 N
Fan Details (Type/Make/Blades)		Type: E-Viscous fan and Size:24"	Type: E-Viscous fan and Size:24"

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8.0 Test result:

8.1 Cruising speed: 40 km/hr & 55 Km/hr

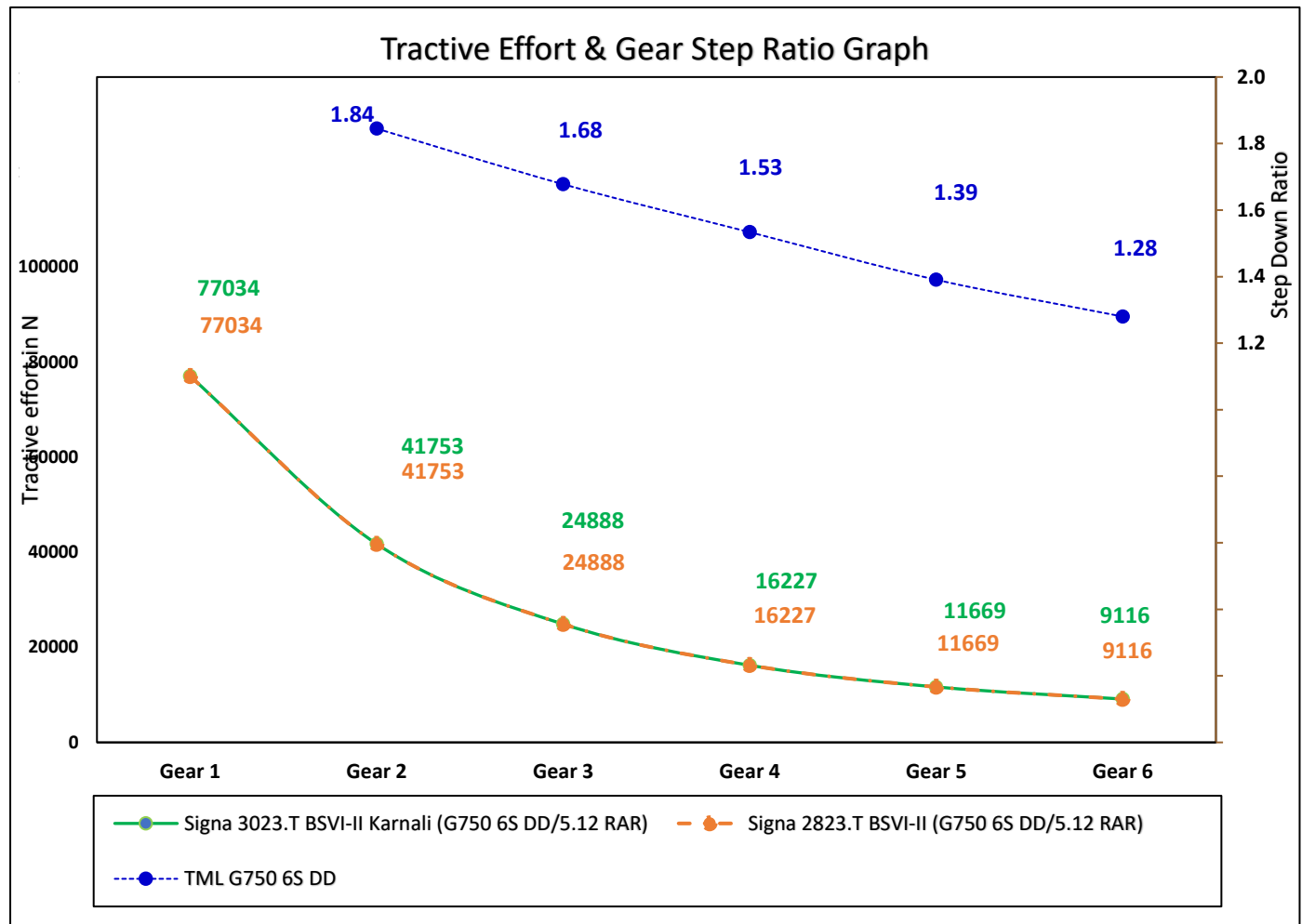
Sr. No.	Parameters			Base Vehicle : OBD-2						Test Vehicle : OBD-2									
				SIGNA 2823.T BSVI-II						SIGNA 3023.T RDE BSVI									
1	Vehicle Identification No. (ERC No. /EJ No./Reg. No.)			ERC-1279						ERC-1219									
2	Vehicle GVW/GCW		kg	28000						30000									
3	Engine Details (Make/cc/Norms/no. of Cyl.)			Cummins BS-VI Ashwamedh 5.6lit						Cummins BS-VI Ashwamedh 5.6lit									
	Engine Max Power			220 HP						220 HP									
	Engine Max Torque			925Nm @ 1000-1300rpm						925Nm @ 1000-1300rpm									
	Engine Calibration Maturity Level			PF10002.12 (Production CAL)						100% Cal Maturity(PF10002.12 (Production CAL))									
4	Regeneration Strategy (Time ase/Soot base)			Time Based (100 Hrs)						Time Based (100 Hrs)									
5	Gear Box (make/drive/FGR)			G750-FGR - 8.45,DD						G750-FGR - 8.45,DD									
	Gear Ratios			1st- 8.45, 2nd- 4.58, 3rd- 2.73, 4th- 1.78, 5th- 1.28, 6th- 1.00, Rev- 7.36						1st- 8.45, 2nd- 4.58, 3rd- 2.73, 4th- 1.78, 5th- 1.28, 6th- 1.00, Rev- 7.36									
6	RAR			5.12						5.12									
7	Tyre Details			All 295/90R 20 (Ceat Make)						All 295/90R 20 (Ceat Make)									
8	LBSC	High & Low Engine Speed Breakpoint		RPM		1900, 1900						1900, 1900							
		Vehicle Mass Threshold		kg		28000						30000							
	VAM	Variable Acceleration 1, 2		kmph/sec		2.5,2.5						2.5,1.6							
		Variable Acceleration Speed 1, 2		kmph		20,60						20,40							
	GDP	GD Max. Veh. Speed (Heavy & Light Engine Load)		kmph		50,50						50,50							
	RSG	Maximum Accelerator Vehicle Speed		kmph		80						80							
	Torque Factor	Torque Limit Medium Adjustment Factor		Balance		100%						120%							
		Torque Limit Maximum Adjustment Factor		Economy		110%						120%							
	PTP	Powertrain Protection (Enable / Disable)				Enable						Enable							
		Torque Limit		Nm		810Nm in 1st and 2nd and 850Nm in rest						810Nm in 1st and 2nd and 850Nm in rest							
	USFE / Torque Table Revision (1/2/3/n)			USFE-1						USFE-1									
9	Fan Details (type/make/blades)			24" E- Viscous fan						24" E- Viscous fan									
FE Trial Details	Route			Jamshedpur-Ranchi-Jamshedpur															
	FE Duty Cycle (Laden/Unladen)			Laden Trial @ 40 kmph		Unladen Trial @ 40 kmph		Laden Trial @ 55 Kmph		Unladen Trial @ 55 Kmph		Laden Trial @ 40 Kmph		Unladen Trial @ 40 Kmph		Laden Trial @ 55 Kmph		Unladen Trial @ 55 Kmph	
	Trial Date			16.02.26	18.02.26	23.02.26	13.02.26	15.02.26	21.02.26	16.02.26	18.02.26	23.02.26	13.02.26	15.02.26	21.02.26				
	Weather Condition (Sunny/Rainy/Cloudy/Windy)			Sunny	Sunny	Sunny	Sunny	Sunny	Sunny	Sunny	Sunny	Sunny	Sunny	Sunny	Sunny				
	Traffic Condition (Light/Average/Heavy)			Average	Average	Average	Average	Average	Average	Average	Average	Average	Average	Average	Average				
	Regeneration (Yes/No)			No	No	No	No	No	No	No	No	No	No	No	No				
	Trip Distance		km	242.0	242.0	242.0	242.0	242.0	242.0	242.0	242.0	242.0	242.0	242.0	242.0				
	Total Diesel Quantity		Litre	58.96	58.48	35.05	64.60	63.00	38.00	61.30	61.10	34.62	67.57	66.43	38.12				
	Total Urea Quantity		Litre	3.60	3.70	1.40	4.00	4.00	2.00	3.80	3.90	1.50	4.40	4.60	2.00				
	Equivalent Urea Quantity : 50% Consumption		Litre	1.80	1.85	0.70	2.00	2.00	1.00	1.90	1.95	0.75	2.20	2.30	1.00				
	Total Eq. Fluid Consumed		Litre	60.76	60.33	35.75	66.60	65.00	39.00	63.20	63.05	35.37	69.77	68.73	39.12				
	Urea uses % wrt. Diesel		%	6%	6%	4%	6%	6%	5%	6%	6%	4%	7%	7%	5%				
	Fuel Economy		kmpl	4.10	4.14	6.90	3.75	3.84	6.37	3.95	3.96	6.99	3.58	3.64	6.35				
	Status of Fuel economy		%	Base	Base	Base	Base	Base	Base	-3.82%	-4.29%	1.24%	-4.40%	-5.16%	-0.315%				
	Equivalent Fluid Economy		kmpl	3.98	4.01	6.77	3.63	3.72	6.21	3.83	3.84	6.84	3.47	3.52	6.19				
	Status of Equivalent Fluid economy		%	Base	Base	Base	Base	Base	Base	-3.86%	-4.31%	1.07%	-4.5%	-5.4%	-0.3%				
	FLE based on Lit/Ton/100 km			0.897	0.890	1.477	0.983	0.959	1.612	0.871	0.868	1.462	0.961	0.947	1.617				
	Status of FLE based on Lit/Ton/100 km			Base	Base	Base	Base	Base	Base	2.9%	2.5%	1.1%	2.2%	1.3%	-0.3%				
	Cluster Total Odo. Reading (at End of trial)		km	244	245	243	245	245	243	244	244	241	244	244	242				
	Cluster Trip AFE		kmpl	4.20	4.40	7.20	4.00	4.00	6.10	3.80	4.10	7.10	3.80	3.80	6.10				
Avg.kmpl Results	Average Fuel Economy		kmpl	4.12	6.90	3.79	6.37	3.95	6.99	3.61	6.35								
	Status of Average Fuel economy		%	Base	Base	Base	Base	-4.1%	1.2%	-4.8%	-0.3%								
	Average Eq. Fluid Economy		kmpl	4.00	6.77	3.68	6.21	3.83	6.84	3.49	6.19								
	Status of Average Eq. Fluid economy		%	Base	Base	Base	Base	-4.1%	1.1%	-5.0%	-0.3%								
	Average FLE by Lit/Ton/100 km			0.89	1.48	0.97	1.61	0.87	1.46	0.95	1.62								
	Status of Average FLE by Lit/Ton/100 km			Base	Base	Base	Base	2.69%	1.06%	1.77%	-0.31%								
Overall Results L+UL	Overall Avg. Fuel Economy (kmpl) : L + UL		80%	20%	4.48	4.13	4.33	3.95											
	Status of Overall Avg. Fuel Economy (%)			Base	Base	-3.40%	-4.23%												
	Overall Avg. Eq. Fluid Economy (kmpl) : L + UL		80%	20%	4.35	4.00	4.20	3.83											
	Status of Overall Avg. Eq. Fluid Economy (%)			Base	Base	-3.45%	-4.41%												

Note: Equivalent Urea consumption is taken as 50% of actual urea consumption.

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9.0 Tractive Effort Comparison :



10.0 Observation: Nil

11.0 Further Testing: Further fluid economy evaluation needs to be done with benchmark vehicle to get the PALS attribute wrt core competitor.

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Annexure-1

Level 1 Attribute:		PANEL Chart		VE CVBU/FE/01
PAT- Fuel Economy		Specification:Max Power:220 hp @ 2300 rpm Max Torque - 925 Nm 1000-1300 rpm Driveline : G-750 DD 5.12 RAR 295/90R 20 Tyres iUSFE Switch Torque Factor: E-120% ; B-120% EOL Parameters : LBSC: 1900,1900 RPM VAM: 2.5,1.6 kmph/s @ 20,40 kmph GDP: 50,50 kmph RSG: 80 kmph PTP: 810N-m & 850 N-m		Panel Chart No: VE CVBU/ HCV/ SIGNA 3023.T 5.6L BSVI-II /Karnali / FE/01
Programme / MY : SIGNA 3023.T Karnali BSVI-II	Gateway : DR3	Owner: Ajaykumar Jadhav	Project Name: SIGNA 3023.T Karnali BSVI-II (Cx Version)	Date: 09/03/2026

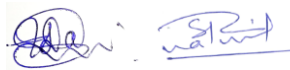
Level 1 Summary:			Statement Of Intent (SOI) or DNA Description:	
Level 1:	PALS Position	Status		Fuel Economy: Haulage Application
		Current/ Gateway	Job#1 Predicted	
	L	UC		

Deriv.	Attribute Target Summary:					PALS	Metric	Target	Status		Base Vehicle		Remarks
	Level 2	Level 3	Level 4	Level 5	Level 6				Current/Gateway SIGNA 3023.T Karnali BSVI-II	Job#1 Predicted	Base Signa 2823.T BSVI-II		
	Specification								LBSC - 1900,1900,30000 Kg VAM- 2.5 Kmph/s, 1.6 Kmph/s, 20 kmph, 40 kmph GDP- 50 kmph, 50 kmph RSG - 80kmph		LBSC - 1900,1900,27500 Kg VAM- 2.5 Kmph/s, 2.5Kmph/s, 20 kmph, 60 kmph GDP- 50 kmph, 50 kmph RSG - 80kmph		
	Vehicle GVW (Payload)						kg		30000		28000		
	Power (Max) @ rpm						HP		220 hp @ 2300 rpm		220 hp @ 2300 rpm		
	Power to Weight ratio						HP/Ton		7.33		7.86		
	Torque (Max) @ rpm						Nm		925 Nm @ 1000-1300 rpm		925 Nm @ 1000-1300 rpm		
	Torque to Weight ratio						Nm/Ton		30.83		33.04		
	Tyres size						--		Ceat 295/90R20		Ceat 295/90R20		
	Gear Box Ratios						--		G750 6S DD, FGR- 8.45		G750 6S DD, FGR- 8.45		
	Final Drive Ratio						--		5.12		5.12		
	Fluid Economy at Rated Load					L			UC				
	Highway AC OFF						--						
	Route : Jamshedpur-Ranchi						--						
	Avg. Fluid Economy @40 kmph						kmpl	≥ 4.00	3.83		4.00	*Signa 3023.T Karnali BSVI-II and Signa 2823.T BSVI-I was tested back to back condition On JSR-Ranchi route.	
	Avg. Fluid Economy @55 kmph						kmpl		3.49		3.68		
	Fluid Economy At Unladen					L			L				
	Highway AC OFF						--						
	Route : Jamshedpur-Ranchi						--						
	Avg. Fluid Economy @40 kmph						kmpl	≥ 6.77	6.84		6.77	*Signa 3023.T Karnali BSVI-II and Signa 2823.T BSVI-I was tested back to back condition On JSR-Ranchi route.	
	Avg. Fluid Economy @55 kmph						kmpl		6.19		6.21		
	Overall Fuel Economy (80% Laden + 20% Unladen)					L			UC				
	Highway AC OFF						--						
	Route : Jamshedpur-Ranchi						--						
	Overall Avg. Fluid Economy @40 kmph						kmpl	≥ 4.35	4.20		4.35	*Signa 3023.T Karnali BSVI-II and Signa 2823.T BSVI-I was tested back to back condition On JSR-Ranchi route.	
	Overall Avg. Fluid Economy @55 kmph						kmpl		3.83		4.00		

Executive Summary:

1. SIGNA 3023.T Karnali BSVI-II & SIGNA 2823.T BSVI-I tested back to back on Jamshedpur-Ranchi route at 40 & 55 kmph cruising speed in laden & unladen condition.
2. Test vehicle SIGNA 3023.T Karnali BSVI-II tested with : 220 HP Engine calibration with 100% Cal Maturity.
3. Base vehicle SIGNA 2818.T BSVI-II tested with production released configuration.
4. Target has been taken at-par (As per Cx approach) wrt to base SIGNA 2823.T BSVI-II vehicle.
5. SIGNA 3023.T Karnali BSVI-II is not meeting the PALS target of "Leadership".

Tested by:


Palash Das / Swapnil Jadhav
PAT ENGINEER

Verified and Approved by:


Ajaykumar Jadhav / Sanjeev Kumar
PAT LEAD / PROJECT LEAD

OUR CULTURE CREDO

AT TATA MOTORS

We are connecting aspirations by being bold in thought and action, owning every opportunity and challenge, Solving together as one team and engaging all our stakeholders with empathy.

We are **MORE WHEN ONE!**

Fuel Economy Evaluation on Signa 3023.T BSVI-II Karnali Vs Signa 2823.T BSVI-II Base vehicle

PED - HCV | 09.03.2026



Vehicle Specification

Sr. No.	Parameters			Base Vehicle	Test Vehicle
				SIGNA 2823.T BSVI-II	SIGNA 3023.T BSVI-II Karnali
1	Vehicle Identification No. (ERC No. /EJ No./Reg. No.)			ERC-1183	ERC-1219
2	Vehicle GVW/GCW	kg		28000	30000
3	Engine Details (Make/cc/Norms/no. of Cyl.)			Cummins BS-VI Ashwamedh 5.6lit	Cummins BS-VI Ashwamedh 5.6lit
	Engine Max Power			180 HP @ 2300 rpm	220 HP
	Engine Max Torque			850Nm @ 1000-1600 rpm	925Nm @ 1000-1300rpm
	Engine Calibration Maturity Level			As per production released	100% Cal Maturity
4	Regeneration Strategy (Time ase/Soot base)			Time Based (100 Hrs)	Time Based (100 Hrs)
5	Gear Box (make/drive/FGR)			G750-FGR -8.45,DD	G750-FGR -8.45,DD
	Gear Ratios			1st- 8.45, 2nd- 4.58, 3rd- 2.73, 4th- 1.78, 5th- 1.28, 6th- 1.00, Rev- 7.36	1st- 8.45, 2nd- 4.58, 3rd- 2.73, 4th- 1.78, 5th- 1.28, 6th- 1.00, Rev- 7.36
6	RAR			4.88	5.12
7	Tyre Details			All 295/90R 20 (Apollo Make)	All 295/90R 20 (Ceat Make)
8	LBSC	High & Low Engine Speed Breakpoint	RPM	1900 rpm,1900 rpm	1900, 1900
		Vehicle Mass Threshold	kg	28000.00	30000
	VAM	Variable Acceleration 1, 2	kmph/sec	2.5,2.5	2.5,1.6
		Variable Acceleration Speed 1, 2	kmph	20, 60	20,40
	GDP	GD Max. Veh. Speed (Heavy & Light Engine Load)	kmph	50, 50	50,50
	RSG	Maximum Accelerator Vehicle Speed	kmph	80.00	80
	Torque Factor	Torque Limit Medium Adjustment Factor	Balance	105%	120%
		Torque Limit Maximum Adjustment Factor	Economy	110%	120%
	PTP	Powertrain Protection (Enable / Disable)		Enable	Enable
		Torque Limit	Nm	810Nm in 1st and 2nd and 850Nm in rest	810Nm in 1st and 2nd and 850Nm in rest
	USFE / Torque Table Revision (1/2/3/n)			USFE-1	USFE-1
9	Fan Details (type/make/blades)			24" E-Viscous, Make - Behr , 11 no's blade	24" E- Viscous fan
10	Base Cal Improvement (Cal Maturity %)			As per production released	100% Cal Maturity

Sr. No.	Parameters	Base Vehicle	Test Vehicle
		SIGNA 2823.T BSVI-II	SIGNA 3023.T BSVI-II Karnali
FE Avenues (Engine Level)	Base Cal Improvement (Cal Maturity %)	As per production released	100% Cal Maturity
	Low Viscous Engine Oil (TMGO-10W30)	Yes (10W30)	Yes (10W30)
	FE Switch	iUSFE	iUSFE
	Engine Features (EOL : Enable/Disable)	Enable	Enable
	ATS Optimization	No	Yes
	APM	No	Yes
	Cruise Control	No	No
	Gear based VAM	No	No
	Intelligent mode shifter	No	No
	Moving Mass Estimation	No	No
FE Avenues (Vehicle Level)	Driveline Optimization	NA	Yes
	GSA (Gear Shift Advisor)	Yes	Yes
	MSA (Mode Shift Advisor)	NA	No
	Low CRR Tyres	No	Yes
	Low Viscous rear axle oil (80W90)	Yes	Yes
	Smart Alternator (LIN)	NA	Yes
	THU/Unitised/Low friction bearing	Yes	Yes

Sr. No.	Parameters			Base Vehicle : OBD-2					Test Vehicle : OBD-2						
				SIGNA 2823.T BSVI-II					SIGNA 3023.T RDE BSVI						
1	Vehicle Identification No. (ERC No. /EJ No./Reg. No.)			ERC-1279					ERC-1219						
2	Vehicle GVW/GCW		kg	28000					30000						
3	Engine Details (Make/cc/Norms/no. of Cyl.)			Cummins BS-Vi Ashwamedh 5.6lit					Cummins BS-Vi Ashwamedh 5.6lit						
	Engine Max Power			220 HP					220 HP						
	Engine Max Torque			925Nm @ 1000-1300rpm					925Nm @ 1000-1300rpm						
4	Engine Calibration Maturity Level			PF10002.12 (Production CAL)					100% Cal Maturity(PF10002.12 (Production CAL))						
	Regeneration Strategy (Time ase/Soot base)			Time Based (100 Hrs)					Time Based (100 Hrs)						
5	Gear Box (make/drive/FGR)			G750-FGR - 8.45,DD					G750-FGR - 8.45,DD						
5	Gear Ratios			1st- 8.45, 2nd- 4.58, 3rd- 2.73, 4th- 1.78, 5th- 1.28, 6th- 1.00, Rev- 7.36					1st- 8.45, 2nd- 4.58, 3rd- 2.73, 4th- 1.78, 5th- 1.28, 6th- 1.00, Rev- 7.36						
6	RAR			5.12					5.12						
7	Tyre Details			All 295/90R 20 (Ceat Make)					All 295/90R 20 (Ceat Make)						
8	LBSC	High & Low Engine Speed Breakpoint		RPM	1900, 1900					1900, 1900					
		Vehicle Mass Threshold		kg	28000					30000					
	VAM	Variable Acceleration 1, 2		kmph/sec	2.5,2.5					2.5,1.6					
		Variable Acceleration Speed 1, 2		kmph	20,60					20,40					
	GDP	GD Max. Veh. Speed (Heavy & Light Engine Load)		kmph	50,50					50,50					
	RSG	Maximum Accelerator Vehicle Speed		kmph	80					80					
	Torque Factor	Torque Limit Medium Adjustment Factor		Balance	100%					120%					
		Torque Limit Maximum Adjustment Factor		Economy	110%					120%					
	PTP	Powertrain Protection (Enable / Disable)			Enable					Enable					
		Torque Limit		Nm	810Nm in 1st and 2nd and 850Nm in rest					810Nm in 1st and 2nd and 850Nm in rest					
9	USFE / Torque Table Revision (1/2/3/n)			USFE-1					USFE-1						
	Fan Details (type/make/blades)			24" E- Viscous fan					24" E- Viscous fan						
FE Trial Details	Route			Jamshedpur-Ranchi-Jamshedpur											
	FE Duty Cycle (Laden/Unladen)			Laden Trial @ 40 kmph	Unladen Trial @ 40 kmph	Laden Trial @ 55 Kmph	Unladen Trial @ 55 Kmph	Laden Trial @ 40 Kmph	Unladen Trial @ 40 kmph	Laden Trial @ 55 Kmph	Unladen Trial @ 55 Kmph				
	Trial Date			16.02.26	18.02.26	23.02.26	13.02.26	15.02.26	21.02.26	16.02.26	18.02.26	23.02.26	13.02.26	15.02.26	21.02.26
	Weather Condition (Sunny/Rainy/Cloudy/Windy)			Sunny	Sunny	Sunny	Sunny	Sunny	Sunny	Sunny	Sunny	Sunny	Sunny	Sunny	
	Traffic Condition (Light/Average/Heavy)			Average	Average	Average	Average	Average	Average	Average	Average	Average	Average	Average	
	Regeneration (Yes/No)			No	No	No	No	No	No	No	No	No	No	No	
	Trip Distance		km	242.0	242.0	242.0	242.0	242.0	242.0	242.0	242.0	242.0	242.0	242.0	
	Total Diesel Quantity		Litre	58.96	58.48	35.05	64.60	63.00	38.00	61.30	61.10	34.62	67.57	66.43	38.12
	Total Urea Quantity		Litre	3.60	3.70	1.40	4.00	4.00	2.00	3.80	3.90	1.50	4.40	4.60	2.00
	Equivalent Urea Quantity : 50% Consumption		Litre	1.80	1.85	0.70	2.00	2.00	1.00	1.90	1.95	0.75	2.20	2.30	1.00
	Total Eq. Fluid Consumed		Litre	60.76	60.33	35.75	66.60	65.00	39.00	63.20	63.05	35.37	69.77	68.73	39.12
	Urea uses % wrt. Diesel		%	6%	6%	4%	6%	6%	5%	6%	6%	4%	7%	7%	5%
	Fuel Economy		kmpl	4.10	4.14	6.90	3.75	3.84	6.37	3.95	3.96	6.99	3.58	3.64	6.35
	Status of Fuel economy		%	Base	Base	Base	Base	Base	Base	-3.82%	-4.29%	1.24%	-4.40%	-5.16%	-0.315%
	Equivalent Fluid Economy		kmpl	3.98	4.01	6.77	3.63	3.72	6.21	3.83	3.84	6.84	3.47	3.52	6.19
	Status of Equivalent Fluid economy			Base	Base	Base	Base	Base	Base	-3.86%	-4.31%	1.07%	-4.5%	-5.4%	-0.3%
	FLE based on Lit/Ton/100 km			0.897	0.890	1.477	0.983	0.959	1.612	0.871	0.868	1.462	0.961	0.947	1.617
	Status of FLE based on Lit/Ton/100 km			Base	Base	Base	Base	Base	Base	2.9%	2.5%	1.1%	2.2%	1.3%	-0.3%
	Cluster Total Odo. Reading (at End of trial)		km	244	245	243	245	245	243	244	244	241	244	244	242
	Cluster Trip AFE		kmpl	4.20	4.40	7.20	4.00	4.00	6.10	3.80	4.10	7.10	3.80	3.80	6.10
Avg.kmph Results	Average Fuel Economy		kmpl	4.12	6.90	3.79	6.37	3.95	6.99	3.61	6.35				
	Status of Average Fuel economy		%	Base	Base	Base	Base	-4.1%	1.2%	-4.8%	-0.3%				
	Average Eq. Fluid Economy		kmpl	4.00	6.77	3.68	6.21	3.83	6.84	3.49	6.19				
	Status of Average Eq. Fluid economy		%	Base	Base	Base	Base	-4.1%	1.1%	-5.0%	-0.3%				
	Average FLE by Lit/Ton/100 km			0.89	1.48	0.97	1.61	0.87	1.46	0.95	1.62				
Status of Average FLE by Lit/Ton/100 km			Base	Base	Base	Base	2.69%	1.06%	1.77%	-0.31%					
Overall Results L+UL	Overall Avg. Fuel Economy (kmpl) : L + UL		80%	20%	4.48	4.13	4.33	3.95							
	Status of Overall Avg. Fuel Economy (%)			Base	Base	-3.40%	-4.23%								
	Overall Avg. Eq. Fluid Economy (kmpl) : L + UL		80%	20%	4.35	4.00	4.20	3.83							
	Status of Overall Avg. Eq. Fluid Economy (%)			Base	Base	-3.45%	-4.41%								

Torque Table

110% TF – Base Vehicle

120% TF – Test Vehicle

E-Mode

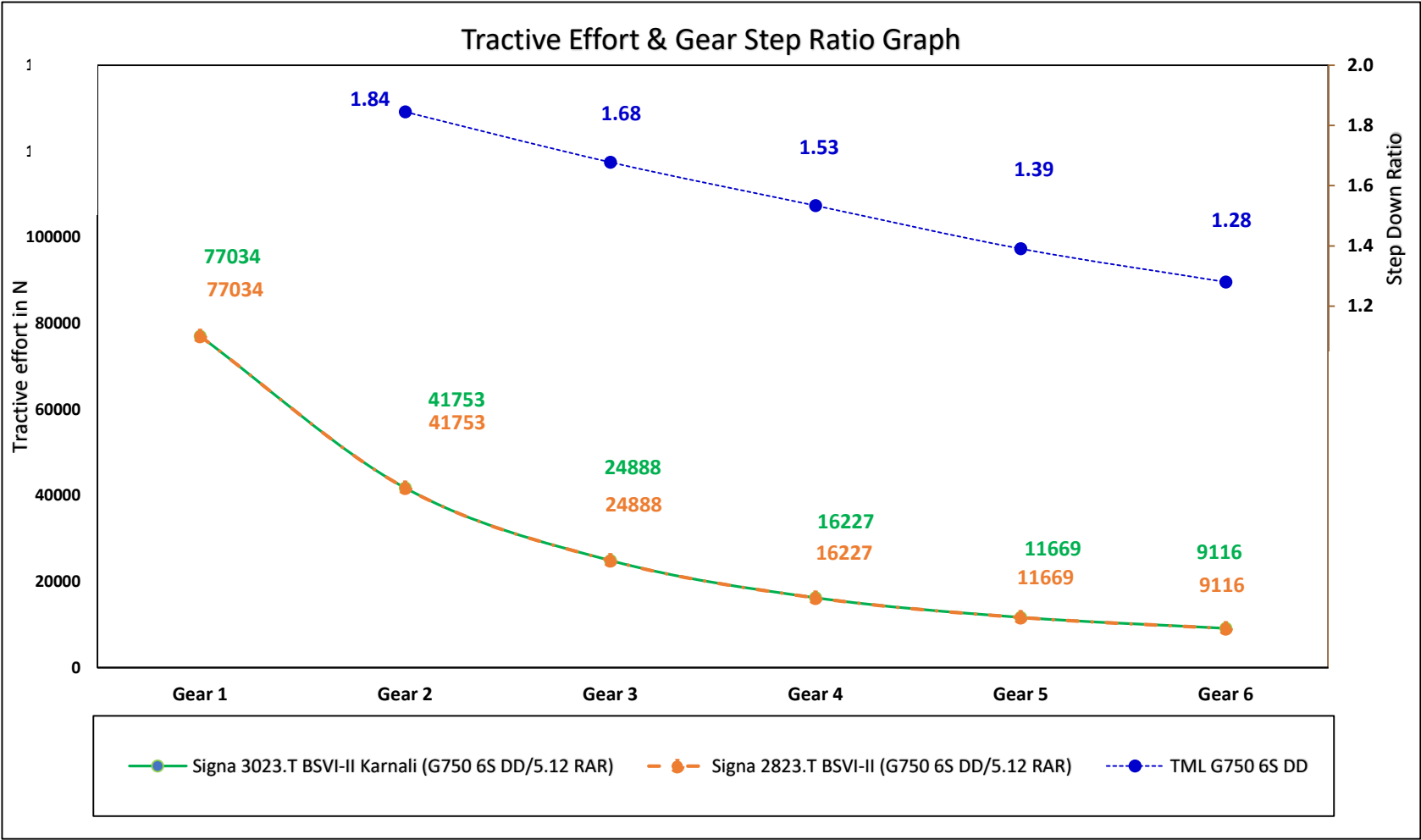
PF10002.12.Signed29.xcal													
C_MCA_USFE_Torque_Lim_Tbl2 × + C_MCA_USFE_Torque_Lim_Tbl1 Screen 2													
Name C_MCA_USFE_Torque_Lim_Tbl2													
C_MCA_USFE_Torque_Lim_Tbl2	X : C_MCA_USFE_GR_X	Y : C_MCA_USFE_EngSpd_Y	Unit : N_m	X-Axis Unit : None	Y-Axis Unit : RPM								
X/Y	600.00	800.00	1000.00	1200.00	1400.00	1600.00	1800.00	2000.00	2200.00	2400.00	2600.00	2650.00	
1.000	570	650	450	450	450	450	400	400	400	400	100	5	
1.328	570	650	450	450	450	450	400	400	400	400	100	5	
1.840	570	650	600	550	550	500	450	450	450	400	100	5	
3.051	570	650	600	550	550	500	450	450	450	400	100	5	
5.199	570	650	925	925	915	850	790	732	692	400	100	5	
9.359	600	650	925	925	915	850	790	732	692	400	100	5	

B-Mode

PF10002.12.Signed29.xcal													
C_MCA_USFE_Torque_Lim_Tbl2 C_MCA_USFE_Torque_Lim_Tbl1 × + Screen 2													
Name C_MCA_USFE_Torque_Lim_Tbl1													
C_MCA_USFE_Torque_Lim_Tbl1	X : C_MCA_USFE_GR_X	Y : C_MCA_USFE_EngSpd_Y	Unit : N_m	X-Axis Unit : None	Y-Axis Unit : RPM								
X/Y	600.00	800.00	1000.00	1200.00	1400.00	1600.00	1800.00	2000.00	2200.00	2400.00	2600.00	2650.00	
1.000	570	650	600	600	600	600	550	550	550	486	100	5	
1.328	570	650	600	600	600	600	550	550	550	486	100	5	
1.840	570	650	650	650	600	600	550	550	500	486	100	5	
3.051	570	650	650	650	600	600	550	550	500	486	100	5	
5.199	570	650	925	925	915	850	790	732	692	486	100	5	
9.359	570	650	925	925	915	850	790	732	692	486	100	5	

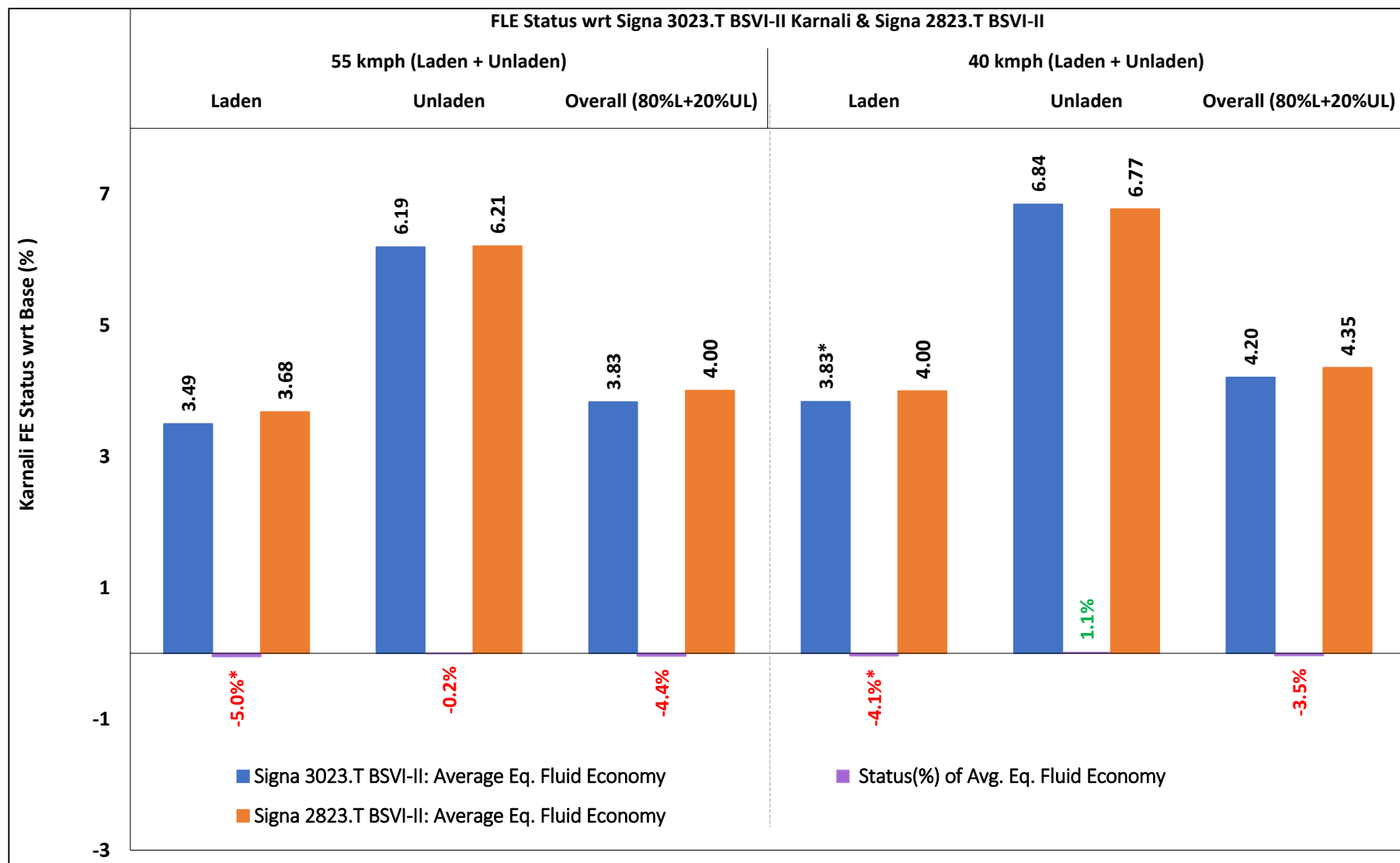
➤ Torque table are same for both Signa 3023.T BSVI-II Karnali (G750) vehicle & Signa 2823.T BSVI-II vehicle(G750).

Tractive Effort Comparison



➤ Tractive effort of Signa 3023.T BSVI-II Karnali (G750) vehicle Vs Signa 2823.T BSVI-II vehicle(G750).

Summary FE Status (Laden & Unladen)



Duty Cycle Analysis Jamshedpur-Ranchi Road FE Trial

Duty Cycle : Summary (Laden)

JSR-RANCHI-JSR Duty Cycle

		16.03.2026 (40 kmph)		15.03.2026 (55 kmph)	
Parameters		Signa 2823.T Karnali Test Vehicle	Signa 2818.T BSVI-II Base Vehicle	Signa 3023.T Karnali Test Vehicle	Signa 2823.T BSVI-II Base Vehicle
Engine idle speed (rpm)		650	650	650	650
Low idle time (%)		5.28	5.59	6.41	7.57
Average engine speed (rpm)		1023.48	1048.56	1222.36	1264.67
Average power (kW)		32.96	31.47	44.18	42.90
Avg. vehicle speed (kmph)	Including Stop	35.84	35.99	43.46	44.28
	Excluding Stop	37.88	38.22	46.51	47.98
FE switch utilization (%time)	Power	21.70	11.86	26.02	23.36
	Balance	0	0	0	0
	Eco	78.30	88.14	73.98	76.64
Gear utilization (%time)	Neutral Gear	5.6	6.1	6.8	8
	4 th Gear	5.6	5.5	6.3	5.7
	5 th Gear	11.52	13.09	8.48	9.10
	6 th Gear	71.88	69.20	73.01	71.03
	Total shifts	279.00	336.00	210.00	253.00
Throttle (%time)	100%	20.6	8.8	24.2	18.9
	0%	40.5	47	41.7	47.6
Coolant Temp (°C)	80°C	0.1	19.7	0.2	12.8
	85°C	1.3	38.2	1.7	36.5
	90°C	51.4	20.8	42.7	26.1
SCRTM Fuel spent (%)		0.0	0.0	0.0	0.0
Regeneration Fuel spent (%)		0.00	0.00	0.00	0.00
Duty Cycle fuel economy (kmpl)		4.42	4.65	4.45	4.05
Duty cycle Distance(km)		249.18	250.20	249.56	249.88
Duty Cycle Time(Hrs)		6.95	6.95	5.74	5.64
Actual Avg. Fuel Economy		3.83	4.00	3.49	4.13

Duty Cycle : Summary (Unladen)

JSR-RANCHI-JSR Duty Cycle

		23.03.2026 (40 kmph)		21.03.2026 (55 kmph)	
Parameters		Signa 3023.T Karnali Test Vehicle	Signa 2823.T BSVI-II Base Vehicle	Signa 3023.T Karnali Test Vehicle	Signa 2823.T BSVI-II Base Vehicle
Engine idle speed (rpm)		650	650	650	650
Low idle time (%)		4.76	4.67	5.90	8.16
Average engine speed (rpm)		976.18	989.75	1177.72	1195.54
Average power (kW)		16.00	17.40	22.44	23.98
Avg. vehicle speed (kmph)	Including Stop	36.43	36.58	42.77	43.35
	Excluding Stop	38.19	38.36	46	47.26
FE switch utilization (%time)	Power	0.27	0.04	0.30	0.08
	Balance	0.00	0.00	0.00	0.00
	Eco	99.73	99.96	99.70	99.92
Gear utilization (%time)	Neutral Gear	4.8	4.9	7.4	8.5
	4 th Gear	2.2	2.1	3.2	2.5
	5 th Gear	2.7	3.5	4.4	4.1
	6 th Gear	86.87	85.64	79.24	78.96
	Total shifts	130.00	165.00	165.00	172.00
Throttle (%time)	100%	0.3	0	0.3	0.1
	0%	23.8	38.6	26.7	40
Coolant Temp (°C)	80°C	0.5	39.2	1.4	17.1
	85°C	1.9	54.8	77.2	77.9
	90°C	88.8	2.2	18	1.9
SCRTM Fuel spent (%)		0.0	0.0	0.0	0.0
Regeneration Fuel spent (%)		0.00	0.00	0.00	0.00
Duty cycle fuel economy (kmpl)		7.91	6.77	6.75	6.70
Duty cycle Distance(km)		245.79	247.18	245.95	247.05
Duty Cycle Time(Hrs)		6.75	6.76	5.75	5.70
Actual Avg. Fuel Economy		6.84	6.77	6.19	6.21

Fluid Economy(FLE) Status of 3023.T BSVI-II Karnali Test vehicle with base 2823.T BSVI-II vehicle :

1. laden

- a) 40kmph :- Fluid Economy of 2823.T is inferior than base vehicle by 4.1 %.
- b) 55kmph :- Fluid Economy of 2823.T is inferior than base vehicle by 5.0 %.

2. UnLaden

- a) 40kmph :- Fluid Economy of 2823.T is superior than base vehicle by 1.1 %.
- b) 55kmph :- Fluid Economy of 2823.T is inferior than base vehicle by 0.2 %.

3. (80% laden + 20 % Unladen)

- a) 40kmph :- Fluid Economy of 2823.T is inferior than base vehicle by 3.5 %.
- b) 55kmph :- Fluid Economy of 2823.T is inferior than base vehicle by 4.4 %.

Glide path for FE improvements :

1) Increase Torque Factor from 120% → 135% (Light mode),

- With the higher TF, **top-gear torque at 1000, 1200 & 1400 rpm** will increase and stabilize at **~608 N·m**.
- This low-rpm torque will reduce the need for Power Mode utilisation of Karnali vehicle having 2 ton extra load.
- Raising TF to **135% in Light Mode** is expected to give a **FE improvement of 2-3%**.

2) Aero Kit Fitment

Vehicle Configuration – 28T Vs 30T - Karnali

ERC No.	ERC 1279/EJ00734 (Base vehicle)	ERC 1219/EJ00659 (Test vehicle)
Vehicle Model	Bebco Cow1 2823.T	Signa 3023.T
Engine No & Shop Order	22H63914054 & 61994	12L63889392 & 61993
Engine Make & Type	Cummins 5.6L - OBD II	Cummins 5.6L - OBD II
Emission Norms	BS-VI OBD-II	BS-VI OBD-II
Max Power, hp	220Hp	220Hp
Max Torque, Nm	925Nm @ 1000-1300rpm	925Nm @ 1000-1300rpm
Cal ID	PF10002.12 (Production CAL)	PF10002.12 (Production CAL)
Vehicle GVW in kg	28000 Kg (As per GVW)	30000 Kg (As per Karnali GVW)
Odometer Reading Before start (km)/Hrs	9989.2 Km/317.1 Hrs	52500.6 Km/1480.4 Hrs
Transmission & Make	G 750DD (FGR:- 8.47)	G 750DD (FGR:- 8.47)
Rear Axle Ratio	41/8 - 5.12	41/8 - 5.12
Powertrain Protection (PTP)	Enable & (810Nm in 1st and 2nd and 850Nm in rest)	Enable & (810Nm in 1st and 2nd and 850Nm in rest)
Engine Low Idle RPM	650	650
Engine High Idle RPM	2600	2600
Idle shutdown time (min)	5 Min	3 Min
Alternator	SEG-5563031_ 24V@100A	SEG-5563031_ 24V@100A
Turbocharger & Starter Motor	5554133 & Lucas TVS_5286815	5554133 & Lucas TVS_5286815
LBSC	1900, 1900 @ 27500 Kg	1900, 1900 @ 30000 Kg
Vehicle mass threshold(kg)	27500 Kg	30000 Kg
VAM	2.5, 2.5 @ 20-60	2.5, 1.63 @ 20-40
GDP	50, 50	50, 50
E-Mode (iUSFE)	110%	120%
Fan control logic	24'INCH BLACK_9_BLADE MAHLE	24'INCH WHITE_11_BLADE BORGWORNER
Fan control setting	Enable	Enable
Fan control drive ratio	1.3	1.3
Tyre Details	All tyre CEAT (Old Tires) -295/90R20	CEAT ALL Tyre (New Tires) -295/90R20

JSR - Ranchi



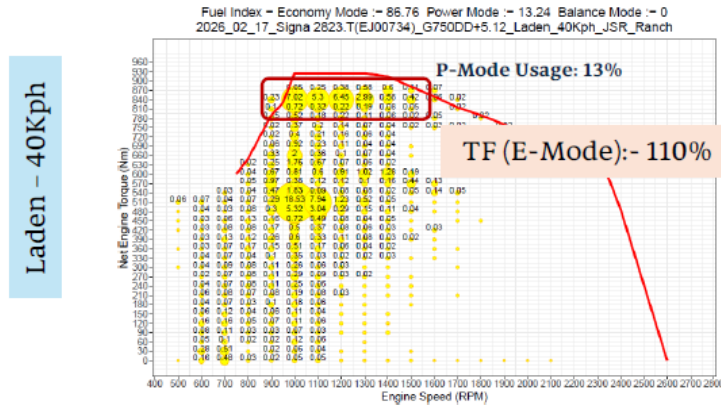
ERC 1279/EJ00734
(Base vehicle)



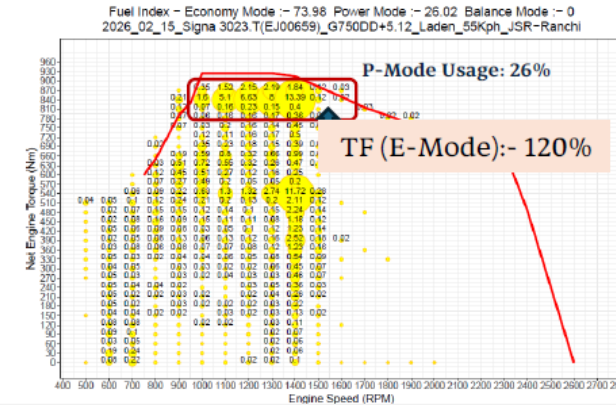
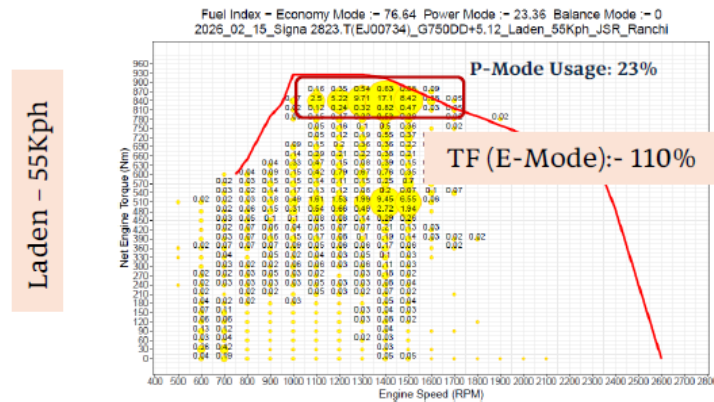
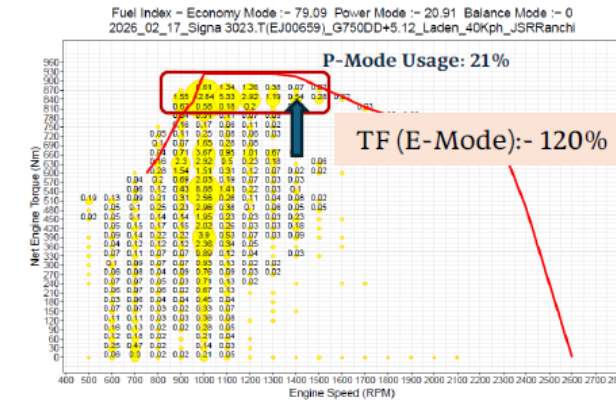
ERC 1219/EJ00659
(Test vehicle)

DC Plots

Fuel Spent % - Signa 2823.T - Base Vehicle



Fuel Spent % - Signa 3023.T - Test (Karnali) Vehicle



Key Take aways:- Despite increasing of TF in Test vehicle (30T) - P-Mode utilisation (%) is higher compared to base vehicle.

Way Forward:- Increasing TF to 135% from 120% for avoiding the P-Mode Usage and Land in lower Torque Zones, New tires were used in test vehicle for FE Trials (To be used old ones/ In-Line with Base vehicle for next set of trials) for reducing overall power consumption.

E-Mode (Base Vs Test Vehicle)

JSR-Ranchi

295 Rev/Km, 5.12 RAR

110% TF – Base Vehicle

X/Y	600	800	1000	1200	1400	1600	1800	2000	2200	2400	2600	2650
1	627	715	495	495	495	495	440	440	440	440	110	6
1.328	627	715	495	495	495	495	440	440	440	440	110	6
1.84	627	715	660	605	605	550	495	495	495	440	110	6
3.051	627	715	660	605	605	550	495	495	495	440	110	6
5.199	627	715	1018	1018	1007	935	869	805	761	440	110	6
9.359	660	715	1018	1018	1007	935	869	805	761	440	110	6

Existing (In Prod.)

120% TF – Test Vehicle

X/Y	600	800	1000	1200	1400	1600	1800	2000	2200	2400	2600	2650
1	684	780	540	540	540	540	480	480	480	480	120	6
1.328	684	780	540	540	540	540	480	480	480	480	120	6
1.84	684	780	720	660	660	600	540	540	540	480	120	6
3.051	684	780	720	660	660	600	540	540	540	480	120	6
5.199	684	780	1110	1110	1098	1020	948	878	830	480	120	6
9.359	720	780	1110	1110	1098	1020	948	878	830	480	120	6

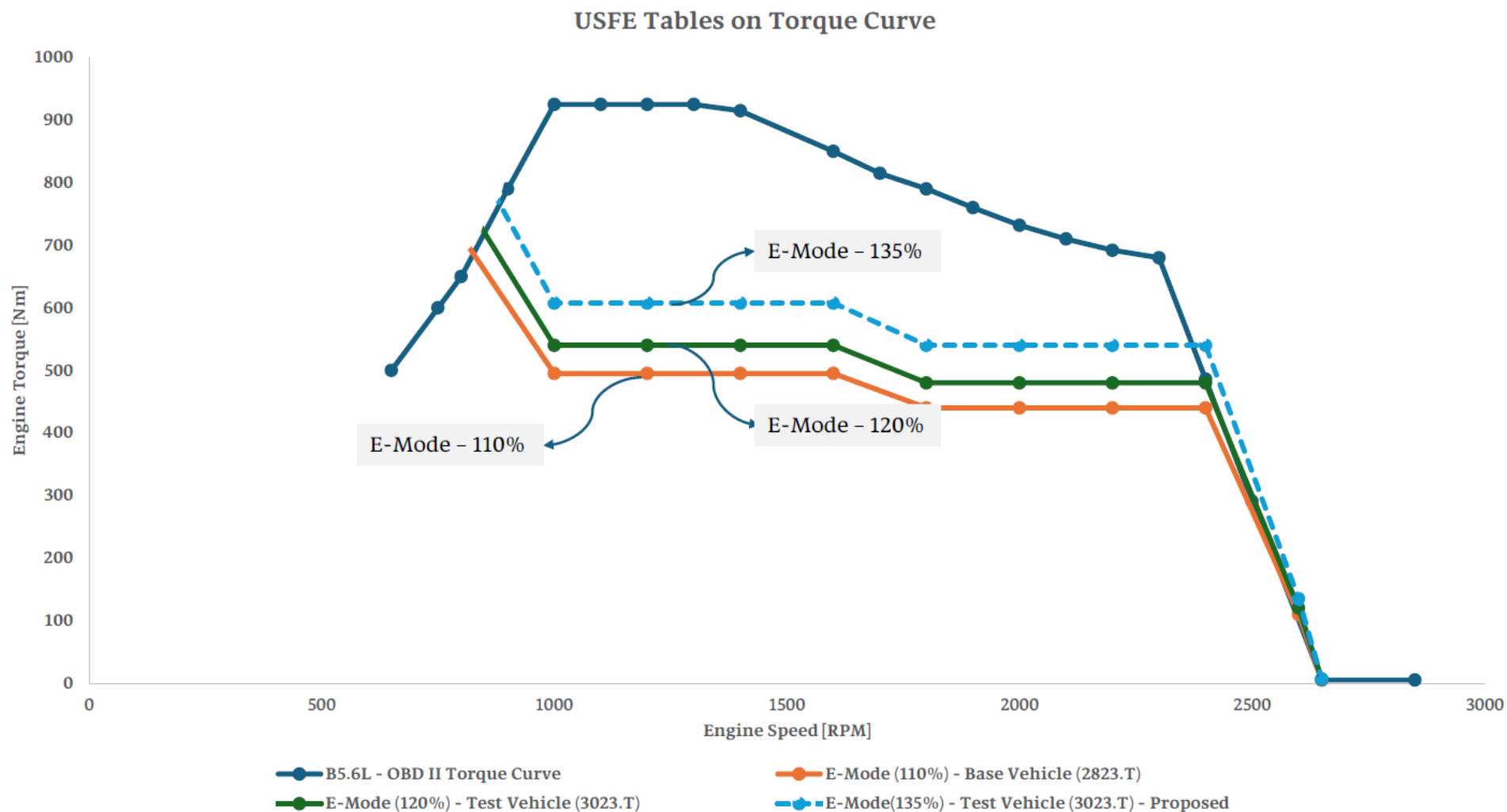
Test Vehicle (Currently)

135% TF – Test Vehicle

X/Y	600	800	1000	1200	1400	1600	1800	2000	2200	2400	2600	2650
1	770	878	608	608	608	608	540	540	540	540	135	7
1.328	770	878	608	608	608	608	540	540	540	540	135	7
1.84	770	878	810	743	743	675	608	608	608	540	135	7
3.051	770	878	810	743	743	675	608	608	608	540	135	7
5.199	770	878	1249	1249	1235	1148	1067	988	934	540	135	7
9.359	810	878	1249	1249	1235	1148	1067	988	934	540	135	7

Proposed as a way forward Testing

E-Mode on Torque Curve



Thank You

TEST REQUISITION

Ref.	SIGNA3023.T/G750/FE/04	Date:	09/03/2026
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To,

ERC Testing

Kindly arrange to take up for testing of the following component.

1	Part Description	TATA SIGNA 3023.T BSVI			
2	Part Number	51810368R,			
3	Drawing number	51810368R	Mod status	A;5,	Mod Date 04/03/2026
4	Vehicle Model / Project	SIGNA 2825.T RDe BSVI			
5	WBS Number	P.21.EA.460TRUCK 3125 / 3725 RDE			
6	Supplier	-			
7	Number of samples	-			
8	Inspection report	Enclosed	✓	If No commitment date:	
9	Material report	Enclosed	✓	If No commitment date:	
10	Suppliers test report	Enclosed	✓	If No commitment date:	
11	Reason for testing	First Source Development	✗	Weight Reduction	✗
		Alternate source development	✗	Cost Reduction	✗
		Data generation Test	✗	Regulatory test	✗
		Any other (pl. specify)	✗	New Development	✓
12	What testing required?				
13	Remarks	FE testing of SIGNA 3023.T BSVI G750 5.12RAR vehicle for increased axle spacing program variant.			
14	Peripheral Parts Availability	No			

For Testing Dept Use only	Test Item Number and Date(stamp)	Test Engineer

Authorized Signature:	
Name:	
Dept:	

Contact Person:	Kundan Kumar
Phone Number	

TATA MOTORS LIMITED ERC PUNE	TEST PLAN	DATE: 13/03/2026			
	VEHICLE TESTING DEPARTMENT				
Description: Report on fluid economy evaluation of Signa 3023.T BSVI-II Karnali Ashwamedh 5.6L (220 HP) engine, G750 (FGR:8.45) Transmission		Test Item No: UA003040			
		New	✓	Moderate	✗
		Major	✗	Minor	✗

Part No: 51810368R,	Modification No: A;5,	Enclosures: Annexure 1-5
Drawing No: 51810368R		
Make: -	WBS Number: P.21.EA.460- TRUCK 3125 / 3725 RDE	Copies to :
Vehicle Type: SIGNA 2825.T RDe BSVI	No. of Samples: -	
Test Requested by:		Equipments:
Test Requisition Reference:	Testing-Outdoor-UA/03069	
Reason for Test/s:	New development	
Regulatory Requirement:	a) Component: AS per work guideline b) Vehicle: AS per work guideline	

Sr. No.	Test description	Test specification	Acceptance criteria	No. of Samples	Test Responsibility	Proto Phase (Mule/ Alpha/ Beta/PP)	Remarks
1	Fuel Economy	AS per work guideline	AS per work guideline	1	Palash Das	beta	completed

REMARKS:	INTERNAL REF:	
TEST TO FAILURE: No		
MEOST: No		
RELIABILITY TESTING: No		
BENCHMARKING TESTING: No		
DFMEA/FMEA DONE: Yes		
PREPARED BY	CHECKED BY	APPROVED BY
Palash Das		

VC No.					VC : 51812168R
Vehicle Model					Signa 3023.T_AC
Parameter	VEPS Name	DEFAULT VALUE *	MIN/MAX *	UNITS	Signa 3023.T Ashwamedh (G-750 DD 15.12 RAR 295/90R20)
Fuel Economy					
Gear Down Protection (GDP)					
Gear Down Protection Enable	20061	N		Y = Enable N = Disable	Y
Gear Down Protection Heavy Load Vehicle Speed	11440	28	Min: 16 Max: (10000)	mph	50 kmph (31.07 mph)
Gear Down Protection Light Load Vehicle Speed	11430	28	Min: 19 Max: 56	mph	50 kmph (31.07 mph)
Idle Speed Control					
Adjustable Low Idle Speed Enable (Read Only)	50512	Y		49	Y
Low Idle Speed	10030	750	Min: 600 Max: 800	rpm	650
Idle Shutdown					
Idle Shutdown Enable	20051	N		Y = Enable N = Disable	Y
Idle Shutdown Timer	10170	60.0	Min: 0.5 Max: 1440.0	Minutes	3
Idle Shutdown in PTO Enable	20071	N		Y = Enable N = Disable	N
Idle Shutdown Manual Override Enable	20081	N		Y = Enable N = Disable	Y
Idle Shutdown service brake switch interaction	50057	N		Y = Enable N = Disable	Y
Load-Based Speed Control					
Load Based Speed Control Enable	50016	N		Y = Enable N = Disable	Y
Load Based Speed Control Low Engine Speed Breakpoint	50027	1800	Min: 1200 Max: 2100	rpm	1900
Load Based Speed Control High Engine Speed Breakpoint	50028	1800	Min: 1500 Max: 2500	rpm	1900
Load Based Speed Control Vehicle Mass Threshold Enable (Read Only)	50446	Y		Y = Enable N = Disable	Y
Load Based Speed Control Vehicle Mass Threshold	50447	68343	Min: 22046 Max: 220460	lb	30000kg (66138.67lbs)
Road Speed Governor					
Road Speed Governor Enable (Read Only)	99001	Y		Y = Enable N = Disable	Y
Maximum Accelerator Vehicle Speed	10000	56	Min: 12 Max: (13590)	mph	80 kmph (49.709 mph)
User Adjustable Fuel Economy					
User Adjustable Fuel Economy Enable	50437	N		Y = Enable N = Disable	Y
UAFE Max Road Speed Enable	50475	Y		Y = Enable N = Disable	Y
UAFE Max Road Speed Limit	50476	36	Min: 12 Max: 56	mph	79 kmph (49.09 mph)
UAFE Torque Limit Enable	50486	N		Y = Enable N = Disable	Y
UAFE Mid Torque Limit Adjust Factor	50487	100	Min: 80 Max: 200	%	120
UAFE Max Torque Limit Adjust factor	50488	100	Min: 80 Max: 200	%	120
Accelerator Control					
Variable Rate Vehicle Acceleration Management	50040	N		Y = Enable N = Disable	Y
Variable Acceleration 1	50041	0.6	Min: 0.0 Max: 8.1	mph/sec	2.5 kph/sec (1.55 mph/s)
Variable Acceleration 2	50042	0.6	Min: 0.0 Max: 8.1	mph/sec	1.6 kph/sec (0.99 mph/s)
Variable Acceleration Speed 1	50043	6	Min: 0 Max: 124	mph	20 kmph (12.43 mph)
Variable Acceleration Speed 2	50044	37	Min: 0 Max: 124	mph	40 kmph (24.85 mph)
Engine Brakes					
Engine Brake Control Enable	23021	N		Y = Enable N = Disable	Y
Engine Brake Type	12790	0		0 = Exhaust Brake 2 = Engine Brake	2
Engine Brake Minimum Vehicle Speed	11540	0	Min: 0 Max: 35	mph	0
Engine Brake Delay Time	11542	0.0	Min: 0.0 Max: 4095.9	Seconds	0
Engine Brake Service Brake Activation Enable	21251	N		Y = Enable N = Disable	N
Engine Brake Clutch Pedal Activation Enable	21253	N		Y = Enable N = Disable	Y
Cruise Control and Engine Brake Interaction Enable	20601	N		Y = Enable N = Disable	N
Accelerator Base					
Clutch Switch	10790	0		0 = Not Installed 1 = Installed	1
Starter Lockout Enable	24000	N		Y = Enable N = Disable	Y
Starter Lockout Relay Type	22161	0		0 = Normally Closed 1 = Normally Open	1
Transmission Setup					
Transmission Top Gear Ratio	10300	1.000	Min: 0.000 Max: 64.000	Ratio	1
Gear Down Transmission Ratio	11340	1.330	Min: 0.000 Max: 64.000	Ratio	1.28
Transmission Type	11410	0		0 = Manual ; 1 = Automatic ; 2 = Fully Automated with Clutch ; 3 = Partially Automated 4 = Fully Automated without Clutch	0
Cruise Control Input					
Cruise Control	20021			Y = Enable N = Disable	Y
Cruise control lower droop	10880			mph	2 kmph (1.242 mph)
Cruise control upper droop	10870			mph	4.9 kmph (3.04 mph)
Cruise control maximum vehicle speed	10090			mph	80 kmph (49.71 mph)
Cruise control auto resume	20611	N		Y = Enable N = Disable	N
Cruise control set speed save	21491	Y		Y = Enable N = Disable	Y
Cruise Control and Engine Brake interaction	20601	N		Y = Enable N = Disable	N
Switch Usage	10100	1		mph	1

Vehicle Speed Input					
Rear Axle Ratio	10050	5.120	Min: 1.000 Max: 15.996		5.12
Tire Size	10060	500	Min: 301 Max: 1287	Rev/Mile	295 rev/km (475 rev/mile)
Number of Transmission Tailshaft Gear Teeth	10070	4	Min: 1 Max: 64		4
Vehicle Speed Sensor Type	10080	0		(0 = None) ; 1 = Magnetic 3 = Data Link - Tailshaft	1
Maintenance Monitor					
Maintenance Monitor Enable	20131	N		Y = Enable N = Disable	Y
Maintenance Monitor Alert Percentage	10860	90	Min: 0 Max: 100	%	99
Maintenance Monitor Operating Mode	10230	0	0=Automatic ; 1=Time 2=Fuel ; 3=Distance	0 = Automatic ; 1 = Time 2 = Fuel ; 3 = Distance	3
Maintenance Monitor Time	10290				1000 sec
Maintenance Monitor Distance	10280	25000	Min: 0 Max: 1000000	Miles	40000 km (24854.848 miles)
Maintenance Monitor Fuel	10250				2404
Alternator Failure Warning	22571	N		Y = Enable N = Disable	Y
Anti-Tampering Measures					
Maximum Engine Speed Without Vehicle Speed Source	10020	2850	Min: 900 Max: 3000	rpm	2100
Engine Protection					
Engine Protection Shutdown Enable	20091	Y		Y = Enable N = Disable	Y
Service Brake					
Service Brake Switch Enable	50070	N		Y = Enable N = Disable	Y
Fan Control					
Air Conditioning Pressure Switch Input	21712	Disable		Y = Enable N = Disable	Y
Min Fan ON Time for Air Conditioning Pressure Switch	11350			Seconds	5
Fan Type	12490				7
Fan drive ratio	50670			ratio	1.3
Air Condition Control (Applicable only for SIGNA cabin)					
Air Conditioner Control	51260				Y
Air Conditioner On Idle Speed Adjust	51262				N
Powertrain Protection (PTP)					
Powertrain Protection	21781	Disable		Y = Enable N = Disable	Y
Driveshaft / Axle Torque Limit	12280	44395		ft-lb	44395 Nm(32744 ft-lb)
Limit 1 Gear Ratio 1	12830	1			0.5
Limit 1 Torque Limit 1	12310	4			850 Nm(626.92 ft-lb)
Limit 2 Gear Ratio 1	12840			ft-lb	4.01
Limit 2 Torque Limit 2	12860			ft-lb	850 Nm(626.92 ft-lb)
Limit 3 Gear Ratio 1	12850			ft-lb	10
Limit 3 Torque Limit 3	12870				810 Nm(597.42 ft-Lb)
Maximum Torque at Zero	12880				850 Nm(626.92 ft-lb)